MILITARY SPECIFICATION SHEET

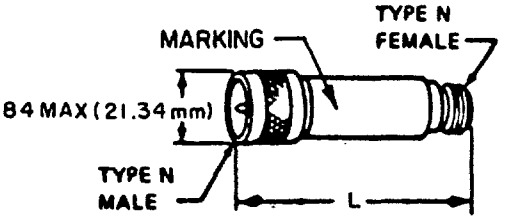
ATTENUATORS, FIXED, COAXIAL LINE (SERIES N),

FREQUENCY RANGE: DC TO 12.4 GHz, AND DC TO 18 GHz,

CLASS III, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the attenuators described herein shall consist of this specification and the latest issue of MIL-A-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

FSC 5985

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ENGINEERING DATA:

Class III.

Operating frequency range: DC to 12.4 GHz and DC to 18 GHz.

Test frequencies for VSWR and attenuation measurements: DC, 4, 8, and 12.4 GHz for dash numbers 01 thru 20; DC, 4, 8, 12.4, and 18 GHz for dash numbers 21 thru 24.

Basic test frequency: 8 GHz.

Operating temperature range: -55°C to +125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Nominal impedance: 50 ohms.

Attenuation: See table I.

Power: See table I.

VSWR: See table I.

Stability of attenuation: See table I.

Terminal connectors: The series N connectors shall meet the interface dimensions of MIL-C-39012. There shall be a male plug on one end and a female plug on the other end.

Part number: M3933/18- (and dash number from table I).

Supersession data: See table III.

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TABLE I. Electrical characteristics.

Dash number	Attenuation dB Nominal	Attenuation dB Deviation DC	Attenuation dB Deviation DC to 12.4 GHz	Attenuation dB Deviation DC to 18 GHz	Maximum power input AV (W) 0 25°C (continuously) 2/	Maximum power input Peak 3/ (kW)	VSWR 1/ Maximum DC to 4.0 GHz	VSWR 1/ Maximum 4.0 to 8.0 GHz	VSWR 1/ Maximum 8.0 to 12.4 GHz	VSWR 1/ Maximum 12.4 to 18.0 GHz	Stability of attenuation Temp change dB/dB/C°	Stability of attenuation Temp cycling dB	Stability of attenuation VIB/ shock dB	Stability of attenuation Moisture resistance dB	Stability of attenuation Peak power dB	Weight (oz) Max	Dimension L (inches) Maximum
01 N S	3	±0.1	±0.2	---	2	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	5	2.70 (68.58)
02 N S	6	±0.1	±0.2	---	2	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	5	2.70 (68.58)
03 N S	10	±0.2	±0.4	---	2	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	5	2.70 (68.58)
04 N S	20	±0.2	±0.4	---	2	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	5	2.70 (68.58)
05 N S	30	±0.3	±0.8	---	2	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	5	2.70 (68.58)
06 N S	40	±0.4	±0.8	---	2	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	5	2.70 (68.58)
07 N S	50	±0.4	±1.25	---	2	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	5	2.70 (68.58)
08 N S	60	±0.4	±1.25	---	2	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	5	2.70 (68.58)
09 N S	1	±0.2	±0.4	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
10 N S	2	±0.2	±0.4	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
11 N S	3	±0.15	±0.3	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
12 N S	5	±0.15	±0.3	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
13 N S	6	±0.15	±0.3	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
14 N S	10	±0.2	±0.4	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
15 N S	13	±0.3	±0.5	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
16 N S	20	±0.3	±0.5	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
17 N S	30	±0.4	±0.8	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
18 N S	40	±0.4	±0.8	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB				Maximum power input		VSWR 1/ Maximum				Stability of attenuation					Weight (oz) Max	Dimension L (inches) Maximum
	Nominal	Deviation															
		DC	DC to 12.4 GHz	DC to 18 GHz	AV (W) @ 25°C (continuously)	Peak 3/ (kW)	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.0 GHz	12.4 to 18.0 GHz	Temp change dB/dB/C°	Temp cycling dB	VIB/ shock dB	Moisture resistance dB	Peak power dB		
19 N S	50	±0.5	±1.0	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
20 N S	60	±0.5	±1.0	---	5	1.0	1.15:1	1.20:1	1.25:1	---	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
21 N S	3	±0.15	---	±0.3	2	0.2	1.12:1	1.15:1	1.18:1	1.20:1	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
22 N S	6	±0.15	---	±0.3	2	0.2	1.12:1	1.15:1	1.18:1	1.20:1	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
23 N S	10	±0.3	---	±0.5	2	0.2	1.12:1	1.15:1	1.18:1	1.20:1	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)
24 N S	20	±0.3	---	±0.5	2	0.2	1.12:1	1.15:1	1.18:1	1.20:1	0.0002	±0.05	±0.1	±0.2	±0.05	3	2.70 (68.58)

1/ VSWR value is for both ends.
2/ Power input is derated linearity to 0 watts at +125°C.
3/ Peak power for a duty cycle of $5X10^{-4}$; maximum pulse duration of 5 microseconds.

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NOTE: Attenuators covered by this specification sheet are substitutable for the manufacturer's or user's part numbers specified in table II. This information in no way implies that manufacturer's part numbers are suitable as a substitute for the military part number.

TABLE II. Substitution data.

Military part number M3933/18-	Manufacturer's code	Manufacturer's part number	NSN
01	99899	774-3	5985-00-135-4056
01	99899	757C-3	5985-00-222-0465
01	99899	757-3	5985-00-814-8383
02	99899	757C-6	5985-01-079-6820
02	99899	774-6	5985-00-135-4057
02	99899	773-6	5985-00-134-9902
02	99899	757B-6	5985-00-118-7985
03	34078	218-10	5985-01-036-6214
03	99899	774-10	5985-00-135-4054
03	99899	773-10	5985-00-231-3195
03	99899	757C-10	5985-00-483-4000
04	99899	757C-20	5985-00-121-4979
04	99899	774-20	5985-00-135-4055
05	99899	773-30	5985-00-115-7974
05	99899	757C-30	5985-00-236-4476
07	99899	757C-50	5985-00-075-8391
08	99899	757B-60	5985-00-106-4256
09	99899	757C-1	5995-00-058-6185
10	34078	218-2	5985-00-163-3985
11	93459	1A-3	5985-00-328-0631
11	93459	1-3	5985-00-498-5838
11	99899	777C-3	5985-00-489-2167
12	93459	1-5	5985-01-157-2805
12	99899	757C-5	5985-01-037-1725
13	93459	1-6	5985-00-110-0344
13	93459	1A-6	5985-00-487-3866
13	99899	777-6	5985-00-267-3030
13	99899	777C-6	5985-00-403-4136
14	93459	1-10	5985-00-411-3698
14	99899	777C-10	5985-00-489-2165
15	93459	1-13	5985-00-275-0415
16	93459	18-20	5985-00-369-7929
16	93459	1-20	5985-00-054-4747
16	93459	1-20N	5985-00-454-6872
16	93459	1-20	5985-00-222-9037
17	93459	1-30	5985-00-110-0343
18	93459	1-40N	5985-00-454-3080
19	93459	1-50	5985-01-053-4164
19	99899	777C-50	5985-00-439-0548
20	93459	2-60	5985-00-759-8893

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TABLE II. Substitution data - Continued.

Military part number M3933/18-	Manufacturer's code	Manufacturer's part number	NSN
21	34078	219-3	5985-01-049-0152
21	34078	389-3	5985-01-152-9026
21	99899	779-3	5985-01-045-8471
22	34078	389-6	5985-01-154-3198
23	34078	389-10	5985-01-152-9027
23	99899	779-10	5985-01-045-5360
24	99899	779-20	5985-01-021-1595
24	34078	219-20	5985-01-049-0151

TABLE III. Supersession data.

Part number M3933/18	Supersedes M3933/	
01	12-1	13-1
02	12-2	13-2
03	12-3	13-3
04	12-4	13-4
05	12-5	13-5
06	12-6	13-6
07	12-7	13-7
08	12-8	13-8

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 99
DLA - ES

User activities:
Navy - MC, SH, AS

Agent:
DLA - ES

Preparing activity:
Navy - EC

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